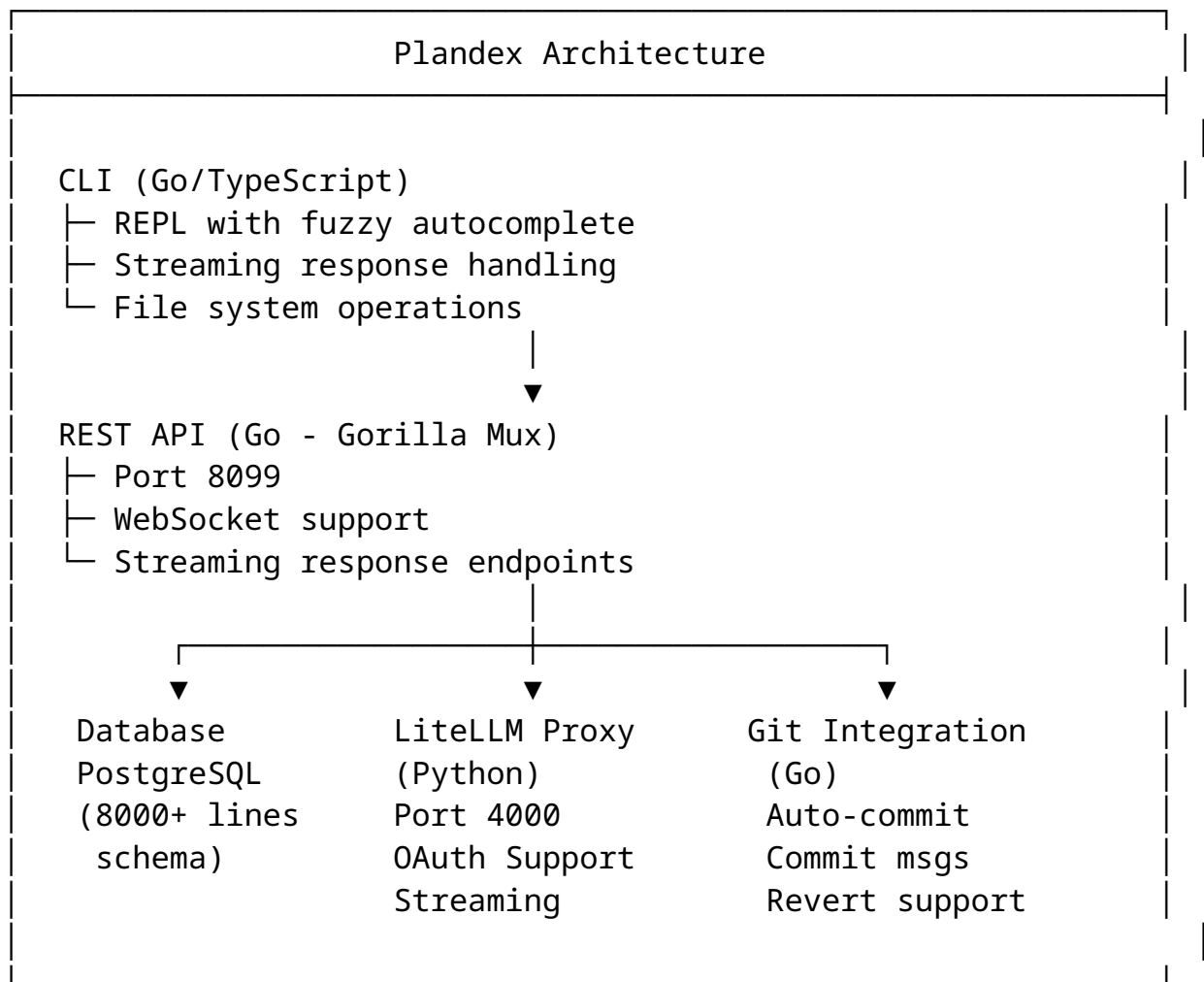


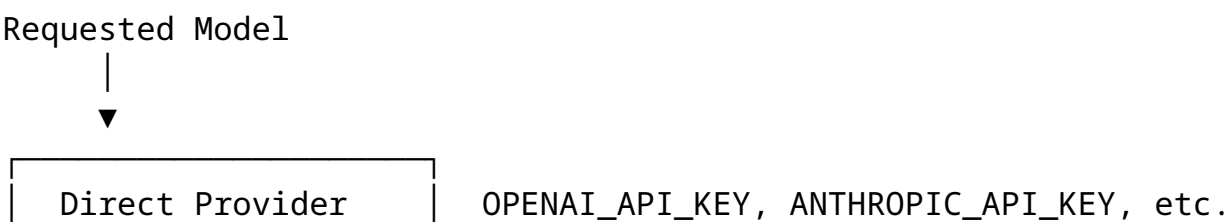
PLANDEX_QUICK_REFERENCE

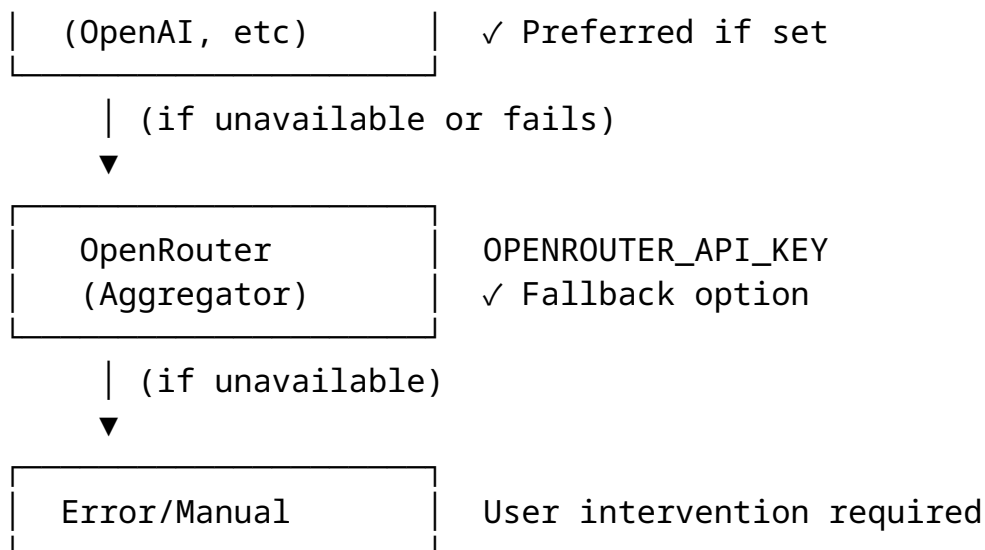
Plandex Quick Reference for HelixCode Porting

Architecture at a Glance



Provider Fallback Chain

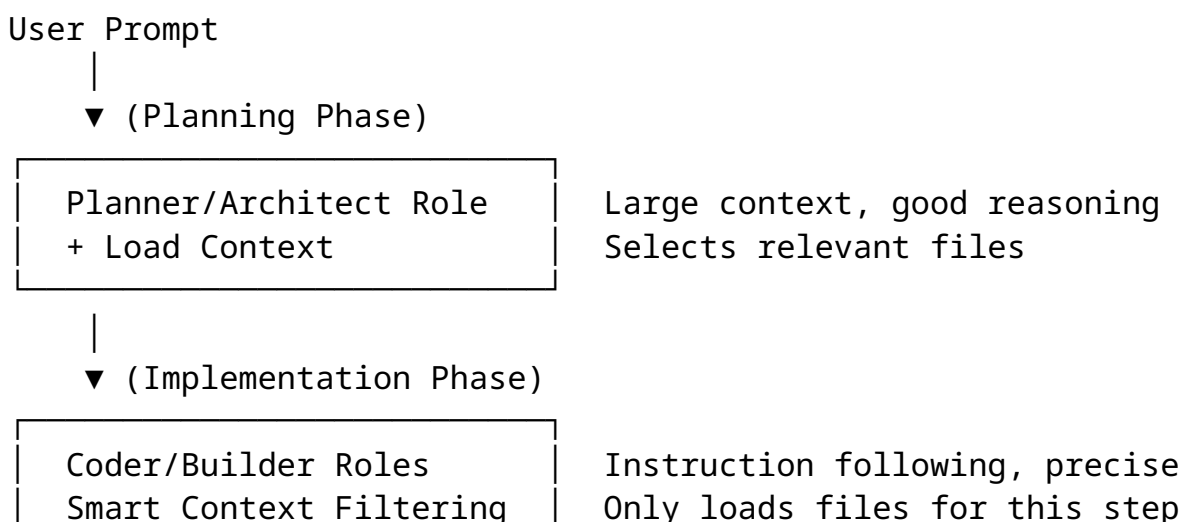


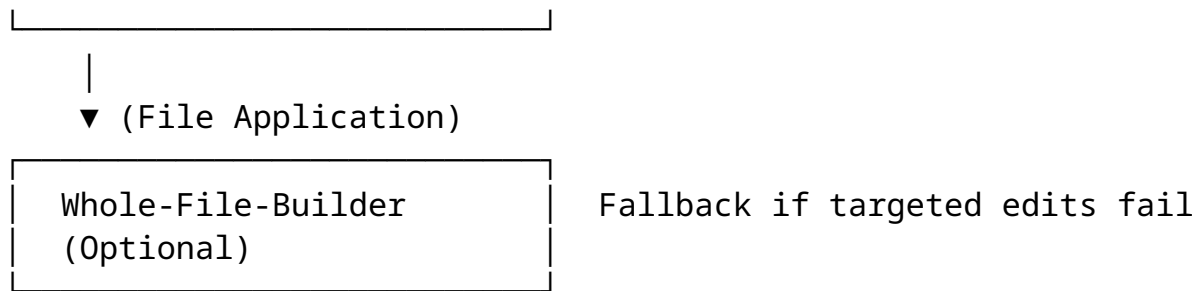


Core Features Matrix

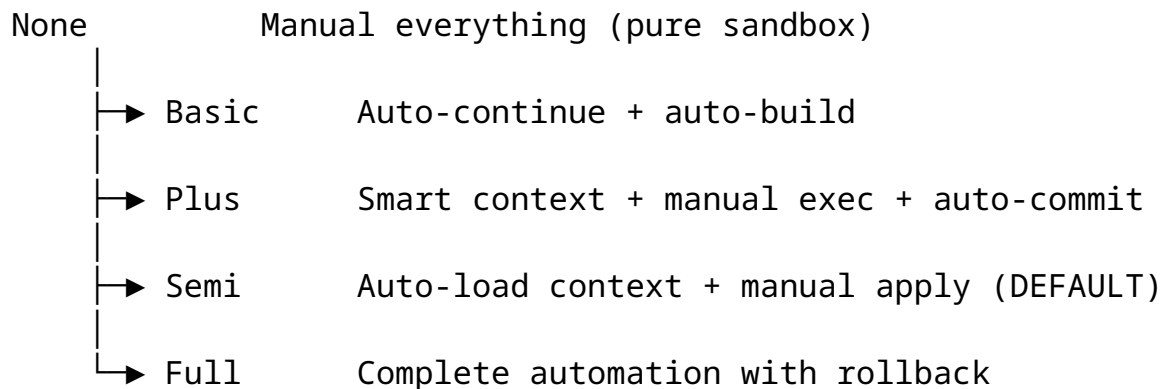
Feature	Implementation	Key Files
Plans	Conv.-like tasks	server/db/plan_helpers.go
Context	Files + URLs + Images	server/syntax/file_map/
Diffs	Git-based review	server/diff/diff.go
Roles	Specialized models	shared/ai_models_roles.go
Autonomy	5 levels + flags	shared/plan_config.go
Versioning	Full history	server/db/ (50+ migrations)
Execution	Build/test/deploy	server/model/plan/build_*.go
Debugging	Browser + Terminal	Auto-screenshot capture

Key Model Roles





Autonomy Levels



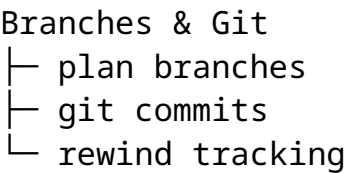
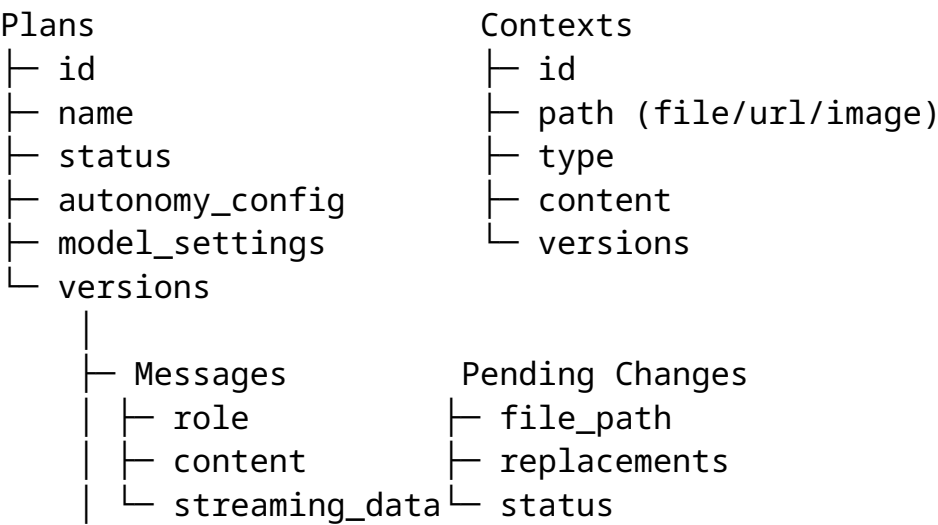
Each level overridable with individual flags: - auto-continue (manual → auto plan continuation) - auto-build (manual → auto file diff building) - auto-apply (manual → auto change application) - auto-exec (manual → auto command execution) - auto-debug (manual → auto error debugging) - auto-commit (manual → auto git commits)

Supported Providers Summary

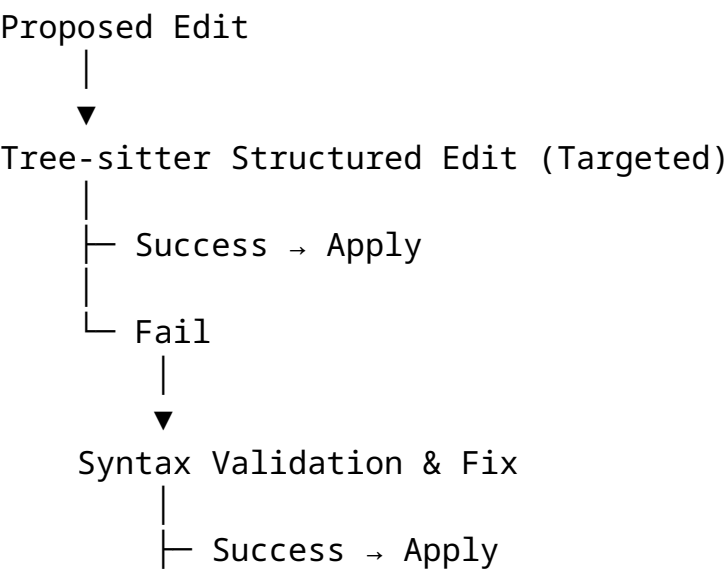
Provider	Type	Auth	Key Features
OpenAI	Direct	API Key	JSON, strict schemas, latest models
Anthropic	Direct	API Key	Claude Max subscription support
Google AI Studio	Direct	API Key	Gemini, free tier available
Google Vertex	Enterprise	Service Account	Bedrock, high volume
Azure OpenAI	Cloud	API Key + URL	Enterprise deployment
AWS Bedrock	Cloud	IAM/Env	Anthropic via AWS
OpenRouter	Aggregator	API Key	100+ models, failover

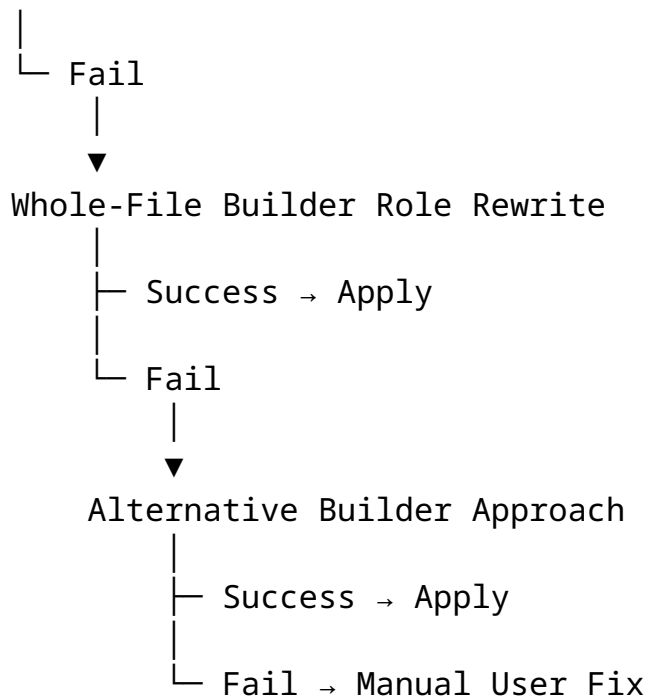
Provider	Type	Auth	Key Features
DeepSeek	Direct	API Key	R1 reasoning models
Perplexity	Direct	API Key	Web-aware reasoning
Ollama	Local	None	Local inference, free
Custom	Self-hosted	Config	OpenAI-compatible endpoints

Database Schema (Essentials)



File Edit Fallback Chain





Context Management Pipeline

1. Discovery
 - └ Tree-sitter project map generation
(30+ languages, 20M+ token support)
2. Selection
 - └ Architect role selects relevant files
using project map + prompt
3. Filtering (Smart Context)
 - └ Load only files relevant to current step
(slides context window up/down)
4. Caching (Optional)
 - └ Apply prompt cache directives for
Anthropic, OpenAI, Google

Configuration Hierarchy

- Preset Autonomy Level
- └ none, basic, plus, semi, full
 - └ Each has standard flags preset
 - └ Individual flags override:
 - └ auto-continue (true/false)
 - └ auto-build (true/false)

- └ auto-apply (true/false)
- └ auto-exec (true/false)
- └ auto-debug (true/false)
- └ auto-commit (true/false)
- └ auto-load-context (true/false)
- └ smart-context (true/false)
- └ auto-update-context (true/false)

Model Pack Overview (16 Built-in)

- General Purpose:** - DailyDriver (default) - balanced - Strong - maximum capability - Cheap - cost optimized - OSS - open source only
- Provider Specific:** - AnthropicOnly, OpenAIOnly, GoogleOnly - OpusPlanner, O3Planner, R1Planner - GeminiPlanner, PerplexityPlanner
- Local/Experimental:** - OllamaExperimental - OllamaAdaptiveOSS - OllamaAdaptiveDaily (hybrid)
- Specialized:** - Reasoning (o3-mini based)

Critical Implementation Files

Area	Files	LOC
DB Schema	server/db/*.go	8,351
Models Config	shared/ai_models_*.go	120K+
Plan Execution	server/model/plan/	~2K
Handlers	server/handlers/	~10K
Syntax/Edits	server/syntax/	~80K
CLI Commands	cli/cmd/	~15K

Credential Discovery (Environment First)

- For each provider:
1. Check specific env var (OPENAI_API_KEY, etc)
 2. For AWS: Check PLANDEX_AWS_PROFILE first
 3. For AWS: Fall back to AWS credential chain
 4. For Google: Check service account file/JSON string/base64
 5. For Anthropic: Support OAuth tokens (sk-ant-oat*)
 6. For Azure: Require both API_KEY and API_BASE

Key Differences from Standard ChatGPT/Claude

Aspect	Plandex	ChatGPT
Context	Project-aware + file loading	Conversation only
Changes	Sandboxed diffs with approval	Direct file writing
Versioning	Full plan history + branching	Conversation history only
Execution	Auto build/test/deploy with rollback	No command execution
Models	Role-based routing	Single model per session
Autonomy	5 configurable levels	None/Full only
Context Size	2M tokens effective	Limited per request
Debugging	Browser + terminal snapshots	None

Quick Start for HelixCode Integration

Phase 1: Foundation (Weeks 1-3)

- ☐ Implement LiteLLM proxy abstraction
- ☐ Set up provider fallback chain
- ☐ Create role-based model routing
- ☐ Store model configs in database

Phase 2: Context & Storage (Weeks 4-6)

- ☐ Build context management pipeline
- ☐ Implement project map generation
- ☐ Create plan version storage schema
- ☐

Build smart context filtering

Phase 3: Execution & Review (Weeks 7-9)

- ☐ Implement diff generation and review
- ☐ Build change application logic
- ☐ Create autonomy level system
- ☐ Add execution tracking

Phase 4: Polish & Integration (Weeks 10-12)

- ☐ Browser debugging integration
- ☐ Git auto-commit functionality
- ☐ Plan branching and rewind
- ☐ Full CLI integration

Environment Variables Quick Reference

Providers

```
OPENAI_API_KEY=sk-...
OPENAI_ORG_ID=org-... (optional)
ANTHROPIC_API_KEY=sk-ant-...
GEMINI_API_KEY=...
AZURE_OPENAI_API_KEY=...
AZURE_API_BASE=...
AZURE_API_VERSION=2025-04-01-preview (optional)
AWS_ACCESS_KEY_ID=...
AWS_SECRET_ACCESS_KEY=...
AWS_REGION=...
PLANDEX_AWS_PROFILE=... (for credential file)
DEEPSEEK_API_KEY=...
PERPLEXITY_API_KEY=...
OPENROUTER_API_KEY=...
```

Google Vertex

```
GOOGLE_APPLICATION_CREDENTIALS=/path/or/json/or/base64
```



```
VERTEXAI_PROJECT=...  
VERTEXAI_LOCATION=...
```

```
# Local
```

```
OLLAMA_BASE_URL=http://localhost:11434
```

```
# Database
```

```
DATABASE_URL=postgres://user:pass@host:5432/db
```

```
# Server
```

```
GOENV=development|production
```

```
LOCAL_MODE=1 (for self-hosting)
```

```
PLANDEX_BASE_DIR=/path/to/data
```